

SHUBHAM KUSHWAH

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🌐 [shubhamKushwah-5](#) | [</> Shubhamkushwah0777 - LeetCode](#) | 📍 Gwalior, India

SUMMARY

Entry-level Backend Developer | Java 21, Spring Boot, MySQL, Redis | Built & Deployed 2 REST APIs (Flash Sale & Trading Journal) | Containerized via Docker with Oracle Cloud (OCI) Deployments | Strong OS (Concurrency, Memory Management), DBMS (ACID, Locks) & Networking Fundamentals | 250+ LeetCode problems | Seeking SDE-1 Role

TECHNICAL SKILLS

Languages: Java 17/21, SQL
Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate
Databases & Caching: MySQL (ACID, Indexing), PostgreSQL, Redis (Lua Scripting)
DevOps & Infrastructure: Docker, Oracle Cloud (OCI), Nginx Reverse Proxy, Git, Postman
Testing & Benchmarking: k6 Load Testing, Concurrency Stress Testing
Core Concepts: Database Locking (Optimistic/Pessimistic), RESTful API Design, JWT, Distributed Systems

EXPERIENCE & PROJECTS

Flash Sale Concurrency Engine | *Java 21, Spring Boot, Redis, MySQL, Docker, OCI, Nginx* [GitHub](#) | [Live Portfolio](#)

- **System Overview:** Engineered a distributed backend system to mimic high-traffic e-commerce flash sales, processing massive surges of checkout requests with a strict zero-oversell guarantee.
- **Cloud Architecture:** Containerized the application via Docker and deployed on Oracle Cloud (OCI). Initially scaled across multiple App Nodes behind a Load Balancer for stress testing, currently consolidated behind an Nginx reverse proxy.
- **Concurrency Optimization:** Replaced standard MySQL pessimistic row-locking (which choked connection pools at 460 req/sec) with an atomic **Redis Lua caching layer** to completely eliminate database lock contention.
- **Peak Performance:** Achieved a peak throughput of **9,317 requests per second (RPS)** with a sub-second p(95) latency of **754ms** by decoupling high-speed cache decrements from asynchronous DB persistence.
- **Extreme Scale Testing:** Validated system resilience using dedicated **k6 distributed load runners**, sustaining **100,000 Concurrent Virtual Users (VUs)** while tuning Linux OS socket limits (`ulimit -n 650000`) to prevent TCP port exhaustion.

Trading Journal API | *Java 21, Spring Boot, Spring Security, JWT, MySQL* [GitHub](#) | [Live Interactive API](#)

- Developed a RESTful trade analytics platform with a stateless security architecture using `OncePerRequestFilter` and JSON Web Tokens.
- Enforced strict cross-user data isolation by extracting authenticated security contexts directly within the service layer, preventing unauthorized resource access.
- Leveraged Java Streams API (`groupingBy`, `mapToDouble`) to aggregate complex portfolio statistics including real-time P&L and strategy win-rates.
- Optimized high-volume trade retrieval by implementing Spring Data Pageable and dynamic sorting for paginated database queries.
- Integrated Apache Commons CSV and multipart file handling to support programmatic data exports and trade screenshot storage.

EDUCATION

Institute of Technology and Management, Gwalior Madhya Pradesh
B.Tech in Computer Science Engineering (CGPA: 7.01) 2021 – 2025

ACHIEVEMENTS

- **Competitive Programming:** Achieved a maximum contest rating of 1650 on LeetCode (Top 20% Globally).
- **Problem Solving:** Solved 250+ Data Structures and Algorithms problems, specializing in Java programming and execution efficiency.